1 The nature of technology: Digital devices

You will learn

- what a digital device is
- the digital devices you can use
- how computers can help you
- things computers cannot do.



In this unit you will learn about digital devices. Devices are things people make to help them. Digital devices have got a computer inside them. The computer makes the devices even more useful.

Did you know?

A computer can check your heartbeat.
The computer uses safe electrical sensors.
A doctor looks at the computer screen.
The doctor can tell if your heart is healthy.



Talk about...

What job would you like to do when you grow up? Will you use computers in the job? How will computers help you?



Learning outcomes: Describe a range of familiar digital devices; Describe tasks where computers can be helpful



Class activation

Some people have jobs that help you. For example, a doctor helps people get better if they are ill.

What other jobs can you think of that help people? Make a class list of these jobs. Next to each job say one way the person might use a computer.

digital device
processor input device
output device touchscreen
hand-held mobile device
tablet computer smartphone





1.1 Digital devices

In this lesson

You will learn:

- what digital means
- what devices are digital.

Spiral back

Last year you learned that the computer uses electricity. In this lesson you will find out more about how a computer uses electricity. You will also learn why a computer is called a digital device.

Switches

A computer works by electricity. Inside the computer there are millions of electrical switches. They are not switches like the light switch. You cannot see them or touch them. They are tiny switches that work by electricity.

The switches inside the computer can be on or off.

- You show an off switch as the number 0.
- You show an on switch as the number 1.

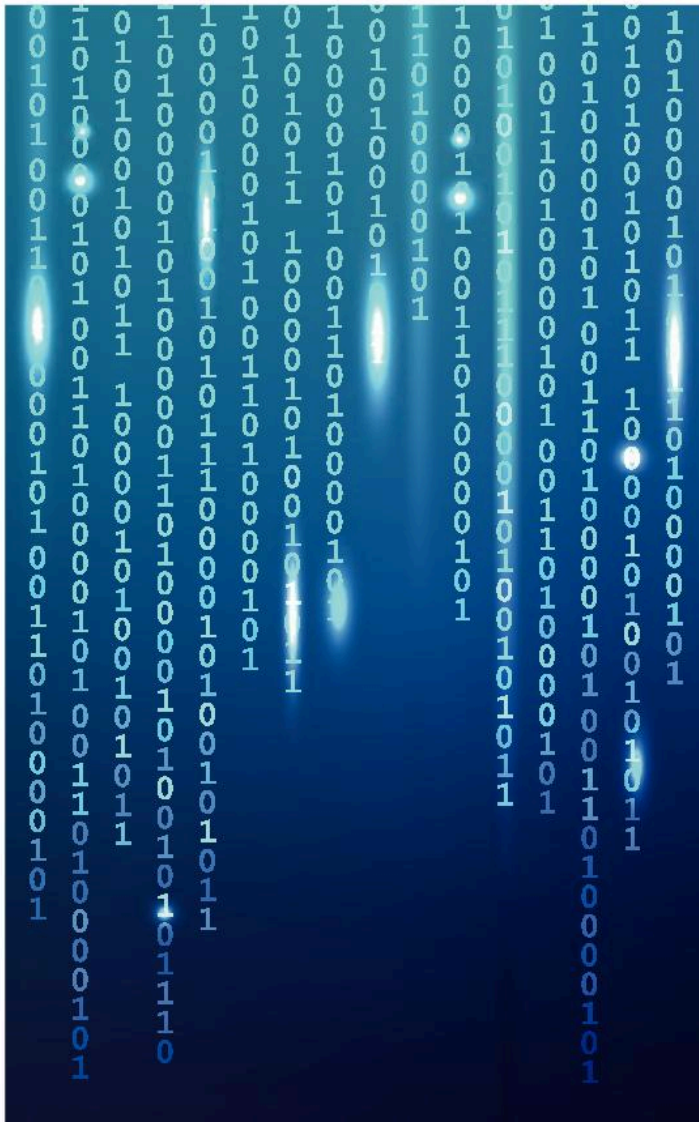
The switches are always changing. Electricity goes through the computer. The flow of electricity turns the switches on and off.

Digital

Computers use the on-off switches to make electronic numbers. That is all there is inside a computer. There are millions and millions of electronic numbers.

Everything inside a computer is made out of those numbers.

Digital means made out of numbers. Everything inside a computer is digital.



Digital device

A device is anything that people make that is useful and helpful.
A **digital device** is any device that has a computer in it.

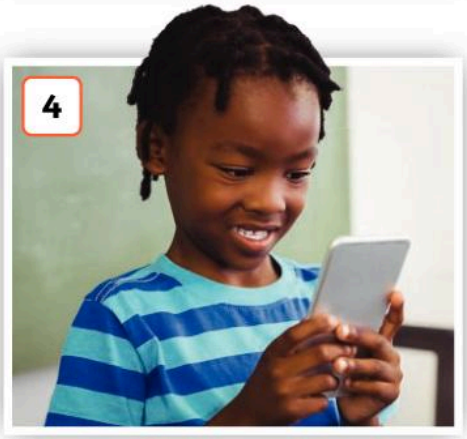
There are many types of digital device. Which ones do you know?

Smartphone | Desktop computer | Tablet | Laptop computer











Activity

Match the devices to the names. Write down the names and numbers.
Or draw the devices and write the names next to the drawing.



Extra challenge

Another type of digital device is called a wearable device. Have you heard of that before? Find out more. Draw or write to show what you found.

Think again

Say something that is not a digital device. Explain your answer.

1.2 The parts of a computer

In this lesson

You will learn:

→ about the parts of a typical computer.

Spiral back



Last year you learned about input and output devices. In this lesson you will learn how they make a computer system.

Processor

Inside every computer is a **processor**. A processor is an electronic device. The processor controls all the other parts of the computer.

Input and output

Other devices are attached to the processor:

- **Input devices** send signals to the processor. Examples of input devices are the mouse and keyboard.
- **Output devices** get signals from the processor. Examples of output devices are the screen and speaker.

The input and output devices let you use the computer.

Desktop computer

In a desktop computer all the devices are quite big. For example, the screen is big. There might be more than one screen.

The devices are separate from each other. They might be joined by wires. They might be wireless.

A desktop computer is big and heavy. It is difficult to carry.



Laptop computer

A laptop computer has smaller parts. The screen and the keyboard are small. All the parts are held in the same case. They are fixed in place.

A laptop is small and light. It is easy to carry.

The touchpad is used instead of a mouse. Some people take a mouse with them and plug it into the laptop.

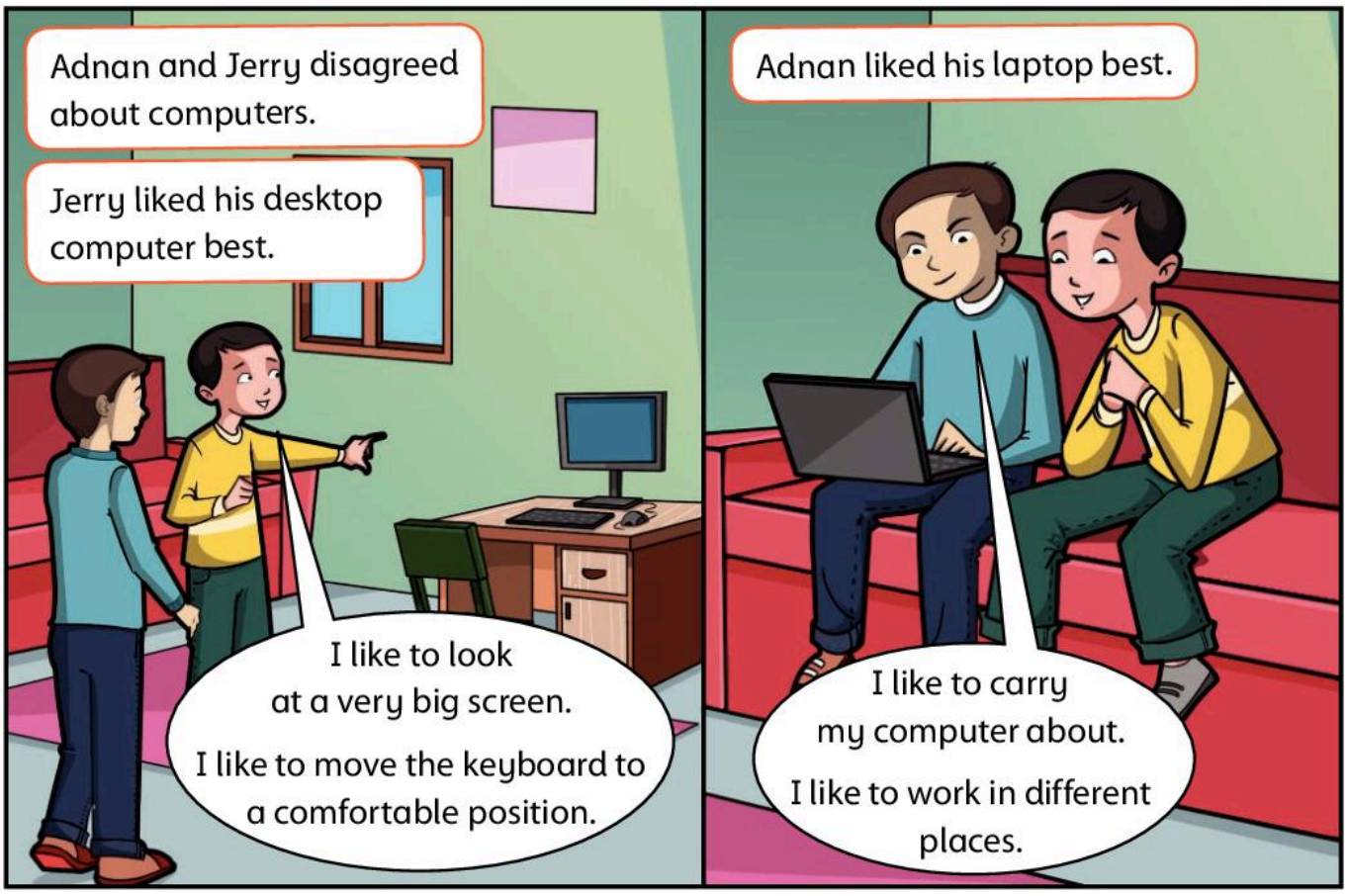


What is a desktop computer used for?

You can use a desktop computer when you want to do lots of work. You can sit and do a lot of typing. You can see your work clearly on a big screen.

Many people like to play games on desktop computers. They like the big screen. They like to use the keyboard and other controls that plug in to the computer.

Which would you choose?



Activity

What type of computer do you use most at school? Draw the computer and label the parts.

Extra challenge

Which do you like best – a desktop or a laptop? Make an advert for the one you prefer.

Think again

What device is at the centre of a computer system? (**Hint:** It is not an input or output device.)